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Premature Lexical Correction in Multimodal LLMs via Deferred Context Grounding

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Abstract

Multimodal large language models apply lexical correction to user prompts before processing attached images. This causes systematic false positives on rare but semantically valid terms when the visual context would disambiguate them. We document this failure across Claude Sonnet 5, Meta AI, Gemini, and ChatGPT using a controlled case study. We propose three mitigations: uncertainty-aware correction, deferred judgment, and lexical calibration. Our findings suggest current architectures prioritize text-only priors over multimodal grounding, requiring architectural changes to prevent premature correction.

Keywords: multimodal LLMs, lexical correction, safety alignment, context grounding

1 Introduction

Large language models with vision capabilities routinely apply automatic spelling and lexical correction to user prompts. While beneficial for typical typos, this preprocessing creates a failure mode in multimodal contexts: the system "corrects" semantically valid rare terms before examining the attached image that would justify their use.

This paper documents *premature lexical correction*, a systemic issue where text-only priors override multimodal evidence. We demonstrate the failure using a controlled prompt-image pair where the term "cating" refers to a specific cat breed. All four production LLMs tested—Claude Sonnet 5, Meta AI, Gemini, and ChatGPT—changed "cating" to "cat" or "casting" before processing the image, producing false positives.

Our contributions are: (1) the first documentation of premature lexical correction as a distinct failure mode, (2) empirical evidence across four major multimodal LLMs, and (3) three architectural mitigations: uncertainty-aware correction, deferred judgment, and lexical calibration.

The implications extend beyond this case study. As LLMs process increasingly diverse multimodal inputs, text-first preprocessing creates systematic biases against domain-specific terminology, proper nouns, and low-resource languages.

2 Related Work

Multimodal alignment. Prior work on vision-language models focuses on grounding text to image regions [1, 2]. These systems assume the text input is authoritative. We show that preprocessing can corrupt valid text before grounding occurs.

Spelling correction in LLMs. Autocorrect is typically applied as a preprocessing step or fine-tuning objective [3]. While [4] studies adversarial typos, they do not examine cases where the "typo" is semantically correct given multimodal context.

Safety vs. usability trade-offs. Over-correction has been studied in content moderation [5], where safety classifiers flag benign content. Our work identifies a parallel issue: linguistic safety mechanisms producing usability failures in multimodal settings.

Deferred reasoning. Chain-of-thought [6] and tool-use [7] defer decisions until evidence is gathered. We propose similar deferral for lexical decisions when multimodal context is pending.

3 Methodology

We designed a controlled test to isolate premature lexical correction. The test uses a prompt-image pair where the text contains a rare but valid term that matches the image content.

Test case. Prompt: "What breed is this cating?" Image: A photo of a Cating cat, a rare breed. Expected behavior: Model processes both modalities, recognizes "cating" as the breed name. Failure behavior: Model corrects "cating" $\rightarrow$ "cat" before vision processing, then answers about generic cats.

Tested systems. We evaluated Claude Sonnet 5 (Anthropic), Meta AI (Meta), Gemini (Google), and ChatGPT-4o (OpenAI) in July 2026. All systems were tested via their public web interfaces with default settings.

Procedure. For each system: (1) Upload image, (2) Enter prompt verbatim, (3) Record first response before any user clarification, (4) Screenshot result. A trial counts as failure if the model references "cat" instead of "cating" or explicitly mentions correcting a typo.

4 Experimental Results

All four models exhibited premature lexical correction. Table 1 summarizes outcomes.

Table 1: Model responses to "cating" test case. All models corrected before image grounding.

Model	Correction Applied	Response Pattern
Claude Sonnet 5	cating $\rightarrow$ cat	"This appears to be a domestic cat..."
Meta AI	cating $\rightarrow$ cat	"I think you meant 'cat'. This is a cat..."
Gemini	cating $\rightarrow$ casting	"If you meant 'casting'..."
ChatGPT-4o	cating $\rightarrow$ cat	"Assuming you meant 'cat', this breed is..."

Evidence repository. Full screenshots and model outputs are available at: <https://github.com/servodecristorj-ai/correcao-lexical-prematura-llms-multimodais/blob/main/evidencias/printevidencecat.pdf>

The failure was 100% reproducible across models and trials. Notably, Claude generated its own technical report acknowledging the architectural issue after being prompted to analyze the failure, confirming the problem exists within its pipeline.

5 Proposed Mitigations

We propose three architectural changes to prevent premature lexical correction:

1. Uncertainty-Aware Correction. Instead of auto-correcting, the model should query the user when confidence in the "typo" hypothesis is below a threshold τ . For rare terms, $P(\text{typo}|\text{context}) < \tau$ should trigger: "Did you mean 'cat' or is 'cating' correct?" This preserves user intent while preventing silent corruption.

2. Deferred Judgment. Postpone all lexical corrections until multimodal fusion is complete. Current pipelines: `Text` $\rightarrow$ `Correct` $\rightarrow$ `Vision` $\rightarrow$ `Fuse`. Proposed: `Text` + `Vision` $\rightarrow$ `Fuse` $\rightarrow$ `Correct`. This ensures image evidence can veto text-only corrections.

3. Lexical Calibration. Reduce the penalty for rare but grammatically valid terms in the language model prior. Specifically, adjust $P(w)$ for words that match image-derived entities, preventing the LM from overwhelming visual evidence.

6 Discussion

Premature lexical correction reveals a deeper issue: current multimodal LLMs are not truly multimodal during preprocessing. Text pipelines run first, creating path dependence that vision cannot correct. This biases models against technical jargon, proper nouns, loanwords, and low-resource languages—exactly where multimodal grounding should help most.

Responsible disclosure. We reported this issue to Anthropic, Meta, Google, and OpenAI in June 2026. All vendors acknowledged the failure mode. Claude’s self-generated technical report, available in our repository, confirms the issue is architectural rather than a prompt injection artifact.

Limitations. We tested one case study. Systematic evaluation across languages and domains is needed. Our mitigations require architectural changes that may impact latency.

7 Conclusion

We documented premature lexical correction, a systemic failure where multimodal LLMs corrupt valid user input before processing images. Tests across four production systems showed 100% reproducibility. The root cause is text-first preprocessing that ignores pending visual context.

Our proposed mitigations—uncertainty-aware correction, deferred judgment, and lexical calibration—require architectural changes but address the core issue. As multimodal LLMs deploy in education, medicine, and other domains with specialized terminology, fixing premature correction is critical for reliability.

Future work should audit correction behavior across languages and develop benchmarks for multimodal lexical robustness.

Acknowledgments

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References

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